

UNIT - III

APPROACHES OF TEACHING

3.1 Approaches of Teaching Biological Science

Science education programmes will be designed to enable the learner to acquire problem solving and decision making skills. Learning of science become unavoidable part of general education. In this chapter we shall discuss various approaches of teaching. Approach is the way in which teaching is approached.

3.1.1. The Concentric Approach

The Concentric approach, often called spiral, is a way of organizing a curriculum by laying out basic concepts, covering other related material, and then circling back around to the basic concept and filling in more complexity and depth.

It is believed that starting with fundamentals that are then regularly revisited, built on, deepened and broadened each time leads to a better understanding of a subject's inter connection and enter into long-term memory.

The concept of concentric design is based on the cognitive psychological theories of Jerome Brunner. Brunner believed that there are three distinct stages in the human Cognitive process; the enactive phase in which the learner interacts with and uses objects or processes; the iconic phase in which the learner manipulates images of these objects or processes; and the symbolic phase, in which abstract representations of them can be utilized. Concentric curriculum design attempts to leverage this understanding of cognition into a deeper understanding of the subject at hand.

Design of Concentric Approach

'Learning spiral' template designed to help educators apply concentric theory to their curriculum design. The template suggests a five phase analysis - learning by getting ready, learning from sources, learning by doing, learning from feedback and learning by thinking ahead that helps generate thinking - centred lessons.

Concept of Concentric approach

- In concentric approach, knowledge is widened slowly and steadily.
- A subject matter is taken and spread over number of years.
- An introductory years, elementary knowledge is given.
- Then in next year little higher level knowledge is added and this is done in all subsequent years.
- Thus a study programme is maintained to impart knowledge to student.

Steps in Concentric Approach

- First a unit / Topic is taken.
- Then the topic is broken into number of sub topics / units.
- These sub topic / units are allotted in various classes according to difficulty level.
- Elementary knowledge is given in introductory years.
- Full-fledged knowledge is provided at higher levels.
- At every stage revision is done.
- Portion is neither too long nor too short.

Merits of Concentric Approach

1. It is a Psychological approach of teaching.
2. Teacher can get adequate time to cover the portion.
3. Topic is continued for number of years, impression lasts forever in minds of learners.
4. As revision is done every year, child retains the knowledge.
5. Since small portion of topic is taken, interest of students is maintained. Teaching does not become dull.
6. Teacher can make use of different techniques to introduce and deliver the content.
7. If same teacher takes topics continuously every year, he/she knows weak points of students and he/she can bring necessary modifications in topics/teaching, to make content strong in students.

Demerits of Concentric Approach

1. If topic is too long, every year, studying the same topic and seeing same chapters in textbook make students loose their interest.
2. If topic is too small, it will not create any permanent impression on mind of learners.
3. If topic is not divided properly, into its sub units, it may prove harmful.
4. Does not work well for small topics.

3.1.2. The Topical Approach

In this approach, the Curriculum maker takes a particular topic as a central theme at various instructions and concepts. A specific topic is started in a particular class and finished in the same class. A topic is a comprehensive

collection of materials of a specific topic of the particular subject.

In this method a particular topic is started in a particular standard and finished over there only. The same topic must not be touch in other grade. In topical approach a topic is taken as unbreakable unit.

Procedure

Topical method is more a system of arrangement of subject matter than a method of teaching. Its adoption depends on a suitable organization of the syllabus. The topic taught at as stretch, without a break or gap. This gives mastery over the topic. This approach is organized around themes, topics and units of the content.

At each stage the topics vary in accordance with the ability and interest of the students. For example in primary classes, the child may start the study of food, clothing, shelter and means of transportation. In the middle classes more important and more difficult topics are introduced. In the secondary classes, the student may be provided with more and more difficult topics.

In this approach, we can deal with all the aspects of a topic and give an overall view of that particular topic.

Examples of Topical Approach

In the topical approach, units focus on discrete events

- i. World War II
- ii. Indian freedom struggle
- iii. Water
- iv. Oxygen etc.

Advantages of Topical Approach

- It enables the teacher to conduct the content as they want and adapt it to the different needs of the children as per their needs.
- It provides a plan to deal with the material in a logical and rational way.
- It helps the student to understand the facts of their developmental setting.
- It gives a sense of purpose to the student.
- This approach can be adopted according to the age, ability and aptitude of the children.

Disadvantages of Topical Approach

- It is difficult to understand all the portions of the topic.
- It gives large amount of information which makes students confused.
- It is quite difficult to study a topic for a lengthy period of time.
- Students don't get a sound knowledge of the content as its conduct in a general way.

3.1.3. Unit Approach

A unit approach plays an important role in learning in the early childhood because it has many meaningful activities related to each other. These learner understands the whole concept. So a unit is a body of information and experience which leads to significant outcomes in a learner. H.C. Horrison was the founder of unit approach.

Definition :

The combination of goals, concepts, generalizations, objectives, skills, attitudes, activities, teaching methods and evaluation procedures into a long-term plan for dealing with a specific topic has generally been called a unit approach in the early childhood classroom.

The unit approach contains four stages. They are

- i. determine the needs of individual children.
- ii. Construct broad objectives that would appear to meet the child's needs
- iii. decide upon a topic that would appropriately incorporate the objectives.
- iv. Plan activities to meet the objectives.

The child learns by actually experiencing or discovering through different methods and activities, rather than just reading a chapter from a text book. The unit approach refers to the idea of studying a topic as a whole instead of several subjects. For example : most people think of water as a science subject one way to look at water is H_2O - a chemistry subject. But it is also art (a beautiful water fall) history (the Red sea) geography (the location of bodies of water) etc.

A unit study approach takes a topic and integrating with it science, social science, maths, language and fine arts etc. For example a unit study about animals would include

- Reading stories (Literature)
- Writing stories about animals (Creative writing)
- Classification of animals (Science)

- Learning new words such as vertebrate and invertebrate (Vocabulary)
- Finding out which animals live on which continents (Geography)

With Unit Approach, history, geography, art, science, maths can be taught together to all ages.

Advantages of Unit Approach

1. The teacher decides the instructional programme according to needs of children.
2. Learning is more integrated.
3. A variety of activities take care of the individual differences among students.
4. It develops the qualities like self-confidence, persistence etc.
5. The teacher can present subject matter in a systematic, logical and effective manner.

Demerits of Unit approach

1. It is time-consuming method.
2. It is over burdening of teacher.
3. To prepare a unit approach and selecting a topic is not an easy task.

3.1.4. Correlated Approach

A mutual relationship or connection between two or more things is called correlation. Generally correlation design is found at the university level where students need expertise in specific subjects. Correlation simply defines how well two subjects go together.

Correlation design includes Bio - Chemistry which is combining of Biology and Chemistry. Other examples are social psychology, bio-statistics which is biology and statistics. Correlation in teaching indicates a technique which shows the reciprocal relationship between various subjects of the curriculum for making the knowledge concrete and permanent. Teachers show similarities or dependence of one subject on another.

Principles of Correlation

- It should be simple, natural, suited to the nature of the subject and the stage of the pupil's mental development.
- It should be adequate and judicious.
- Main topic or main subject should be the main focus. Other topics / subjects should be linked to it and then revert again to the main topic.
- When doing systematic correlation, teachers could sit together and discuss how to correlate.
- It should involve the previous knowledge of the students.
- It should be done only where it is possible, else it leads to superficial correlation.

Types of Correlation

1. Correlation with practical life : Correlation of the given subject with daily activities.
2. Vertical / Internal Correlation : Correlation between the different branches of a given subject.
3. Horizontal / External Correlation : Correlation between the given subject and other subjects.

1. Correlation with practical life

- According to Herbert spencer, the main aim of education is to prepare students for future life. This aim can be achieved only if education is correlated with life. Therefore teaching of various subjects should be correlated with various aspects of life.
- A subject is best understood when it is applicable to daily life.
- Correlation of a subject with daily life is of the utmost importance in order to create interest in the subject.
- Correctional with daily life makes the subject relevant instead of being only theory with no practical application.

For example

Maths : Apply the formulas to calculate areas of rectangle or square of calculate area of classroom or home.

2. Vertical / Internal Correlation

- This type of correlation indicates the relationship between different branches of a given subject.
- It also includes correlation of different topics in the same branch of a given subject.
- Branches of a subject are taught by different teachers, such that each branch is treated as a different entity.
- Internal correlation is necessary for continuity of knowledge and understanding of the subject.

For example

- A science teacher uses vertical correlation to correlate Physics and Chemistry or chemistry and biology and physics or biology and physics etc.

- An economic teacher uses vertical correlation to correlate production, consumption, distribution, exchange etc.
- A mathematics teacher would use vertical correlation to correlate Arithmetic and Algebra or Algebra and Geometry etc.
- A history teacher would use vertical correlation to correlate political history to economic history or social history etc.

3. Horizontal / External Correlation

- This type of correlation is between different school subjects and a given school subject.
- All subjects of the school curriculum contribute towards the realization of the aims of education.
- Since they have the same purpose, study of one subject helps in the study of other subjects.

Horizontal Correlation examples

- i. Mathematics with other subjects
- ii. Science with other subjects
- iii. History with other subjects etc.

Uses of Correlation

1. Mind perceives knowledge as a whole.
2. Retention of earlier knowledge. Always link old knowledge with new.
3. Correlation of a subject with daily helps to make the knowledge useful.
4. All round development

5. It develops the mental abilities like imagination power, logical thinking and analytic power of students.
6. It strengthens skills
7. It makes learning concrete and permanent.
8. It makes the lesson easy and clear
9. It helps to enable the students to acquire knowledge in a short period of time.
10. Develops various human and social qualities such as co-operation, citizenship etc.
11. It helps to lighten the burden of curriculum.
12. It helps the teacher to complete the curriculum in very short period of time.

3.1.5. Integrated Approach

An integrated approach is designed to focus on learning within the curriculum. It focuses on making connections among concepts and experiences so that information and skills can be applied to new and complex issues or challenges.

The concept behind integrated approach is to make meaningful connections between disciplines and utilize critical thinking to real-life problems.

Characteristics of Integrated Approach

- It primarily focuses on problem-solving
It is a compilation of assessment, curriculum development and faculty development.
- Integrated teaching explores and uses information effectively.

- It enables children to integrate ideas and experiences and apply them to formulate new learning situations.
- Creativity, adaptability, critical reasoning and collaboration are the key feature.
- The method of teaching accommodates a variety of learning styles, theories and multiple intelligences.

Values of Integrated Approach

- It allows students to engage in purposeful relevant learning.
- It encourages students to see interconnectedness and interrelationships between the curriculum areas.
- Students are able to use their prior experiences to construct learning.
- Opportunities for students to display competence rather than relying a written or oral text.

Important points to use when implementing the Integrated approach

- Write the integrated curriculum in the daily schedule and daily or weekly lesson plan.
- Foster an atmosphere that welcomes and encourages creativity in the classroom.
- Use age-appropriate materials and techniques in teaching.

Merits of Integrated Approach

1. Integrated approach pays particular attention to an increase in understanding, retention and application of general concepts.
2. Focuses on basic skills and content

3. It provides a better understanding of the content.
4. Integrated approach encourages active participation in relevant real-life experiences.
5. It serves as a connection between various curricular disciplines.
6. It develops higher-level thinking skills.
7. Ensures active participation of students.
8. It create a positive and collaborative learning environment.
9. Teachers intrinsically motivate students to succeed is real life.
10. Provides deeper understanding to content.

Demerits of Integrated Approach

1. There is not enough time in the day to teach everything in isolation.
2. Lack of time to plan effective units.
3. Teachers are reluctant to put the time and effort to implement something that does not guarantee exceptional results.
4. A successful integrative approach involves input from teachers from all different disciplines which is often a difficult task.

3.2 Steps in planning a lesson

Education should be goal oriented and objective based, otherwise it is like navigating without a magnetic compass to show and guide the direction. A key aspect of effective teaching is having a plan for what you are going to teach in the classroom. Creating such a plan involves setting realistic goals deciding how to incorporate course, text books

and resource materials and develop activities that will promote learning. The following points will help in planning a lesson.

Setting lesson goals

- Building up their personality and character so that they can take their rightful place in the mainstream of the nation.
- Assimilating vocationally and socially desirable knowledge and skills.
- Developing ability to think critically and creatively.
- Developing healthy scientific attitudes towards problems and people and
- Motivating them to act according to worthy ideals in the service of community and the nation.
- Goals are based on the following order
 - ✓ Country
 - ✓ Education System
 - ✓ Educational Institution
 - ✓ Class
 - ✓ Unit
 - ✓ Lesson

All goals must be SMART

S – Specific Can the goal precise and well defined?
Is it clear?
Can every one understand it?

M – Measurable What evidence is needed to confirm the completion?

How you will judge whether the goal have been attained or not?

A – Achievable Is it within your capacity to achieve? Are there sufficient resources are available to achieve?

R – Realistic It is possible for you to perform the goal. How does it fit in to your work?

T – Timed Is there a deadline?

Is it feasible to attain in time?

Example

1. At the end of the lesson students know some general concept about the organisms.
2. At the end of the lesson students know about the functions of the organ system,
3. At the end of the lesson students know to draw the organ and to mark its parts.
4. At the end of the lesson students write about parts of the body and its functions.

3.3. Designing a Lesson Plan

Lesson Plan

“Classrooms determine the India’s future”. Taking the importance of the above statement, lesson plan helps the teacher to plan and prepare classroom teaching. If a teacher goes to class without any preparation, he will not only waste the time but also face some problems. Hence to plan and prepare what to teach in each period, how to present it with

systematic order and evaluation procedures to be done in that class room-all combining and presenting in short is called lesson plan.

Lesson Plan Meaning

In lesson plan teacher has to clearly indicate, what are the teaching objectives and how he is going to teach the content in a specific teaching period. What are the aids to be used, what are the steps to use formative evaluation and what are the assessment are to be given etc. should be planned and noted in lesson plan.

Definitions of lesson plan

Good defines a lesson plan as an "outline of the important points of a lesson arranged in order in which they are to be presented to students by teacher".

Bossing defines a lesson plan as an "organized statement of general and specific goals together with the specific means by which these goals are to be attained by the learner under the guidance of the teacher on the given day".

Stand, Lester B defines "a lesson plan is actually a plan of action. It includes the working philosophy of the teacher, her knowledge of philosophy, her information about the understanding of their pupils, their comprehension of the objectives of education, their knowledge of the material to be taught and their ability to utilize effective methods".

Principles for preparing the lesson plan

While preparing a lesson plan, the following principles should be kept in mind.

1. Student's previous knowledge, their capabilities must be basic for a lesson plan.

2. Present the subject matter in an organized, systematic and effective manner.
3. All objectives of lesson both general and specific should know to the teacher.
4. Lesson plan should create interest in students.
5. Questions motivating self-thinking, though-provoking etc. should find a place in lesson plan.

Need and Important features of a lesson plan

A good lesson have the following important characteristics.

1. All the objectives of the lesson both general and specific should be stated clearly in the lesson plan.
2. A good lesson plan should outline in detail the various steps that the teacher proposes to take in the class.
3. It prevents waste of time, as every step has been planned and written.
4. Lesson plan gives self confidence to the teacher.
5. A good lesson plan should indicate the list of AV aids to be used and should specify when and how to use them while teaching.
6. Sometimes we may give more time to unimportant subject matter. We can rectify this defect while preparing lesson plan.
7. A good lesson plan will integrate previous knowledge and experience to new knowledge.
8. A good lesson plan should stimulate independent thinking, interest, creative, imagination powers of the student.
9. A good lesson plan should be flexible and not rigid.
10. To complete the syllabus in time, lesson plan is needed.

11. It helps the teacher to ask proper and important questions to students.
12. It promotes a healthy learning environment.

Uses of Lesson Plan

1. Lesson plan helps the students to focus on the subject matter.
2. It helps to complete the lesson in time.
3. It prevents waste of time.
4. We can determine which Instructional Objectives have been fulfilled with the help of the lesson plan.
5. To become master in subject matter.
6. To get clarity in teaching activities.
7. To obtain teaching resources
8. While teaching it helps to avoid concept contradictions
9. It helps to prepare appropriate teaching aids
10. It brings self-confidence and self-power to teachers.

3.3.1 HERBARTIAN STEPS

German educator J.F. Herbart advocated five formal steps in teaching. They are

1. Preparation
2. Presentation
3. Association
4. Generalisation
5. Application

1 Introduction / Preparation

The introductory phase is called the 'warm-up' phase. The purpose is to create interest in students in the topic. The teacher may relate the new information to the one already

taught or to the previous knowledge possesses by them. The new information is anchored with the existing knowledge.

2. Presentation

The transaction idea is done step by step following psychological approach. Starting from known things, the teacher leads the students to the new information by using comparisons, analogies, illustrations, demonstrations and other audio-visual aids including the chalk board.

3. Association

Through assimilation of the new idea through comparison with former ideas and consideration of their similarities and differences in order to in plant the idea in the mind.

4. Generalization

In science lessons generally the learning material leads to certain generalization leading to establishment of certain formulae, principles or laws. An effort should be made that the students draw the conclusions themselves. Teacher should guide the students only if their generalization is either incomplete or irrelevant.

5. Application

In this step the knowledge gained is applied to certain situations related to their everyday life. This step is in conformity with the general desire of the students to make use of generalization in order to see for them where the generalizations are valid in some other situations.

3.3.2 Preparation of lesson plan

While preparing a lesson plan generally the following steps are followed.

1. Motivation / Preparation
2. Presentation
3. Association / Comparison
4. Generalisation
5. Application
6. Recapitulation
7. Assignment

1. Motivation / Preparation

Students are encourage and motivate to learn new knowledge. The mind of the child should be prepared to receive new knowledge. Preparation may involve.

- Testing previous knowledge to learn new knowledge.
- Integrating the previous knowledge with the new knowledge.
- Arrest attention of the students using new techniques.
- Announcing the aim of the lesson.

2. Presentation:

Here teaching takes important place. Students learn new knowledge. For an effective learning outcome, teacher should ensure active student participation with the use of audio-visual aids. Teacher should stimulate thinking by asking number of questions while teaching.

3. Association / Comparison

Here the student understand the nature of the problem and compare the given information and analyse for the solution of the problem. Here knowledge is correlated with existing knowledge.

4. Generalization

In Mathematics lessons generally the learning material leads to certain generalization leading to establishment of certain formulae, principles or laws. An effort should be made that the students draw the conclusions themselves. Teacher should guide the students only if their generalization is either incomplete or irrelevant.

5. Application:

Here students apply the rules, formulas that they have learnt to solve the problems. By application of knowledge, the validity of generalization is tested and verified. Thus the knowledge becomes more meaningful to the learner and becomes permanent in his mind.

6. Recapitulation

Recapitulation helps the students to review the subject matter and consolidate ideas. Here Teacher will ascertain how well the students understood the new knowledge by way of asking few questions. From the answers we will also judge the effectiveness of the Teaching method.

7. Assignment / Home work

To learn mathematics effectively, class room activity alone is not enough. To know the mathematical concepts, laws, formulae and applications clearly we need home work along with class room activities. Home work revise the portions done in the classroom. Assignments, motivate and reinforce further learning.

3.4 Organising Teaching

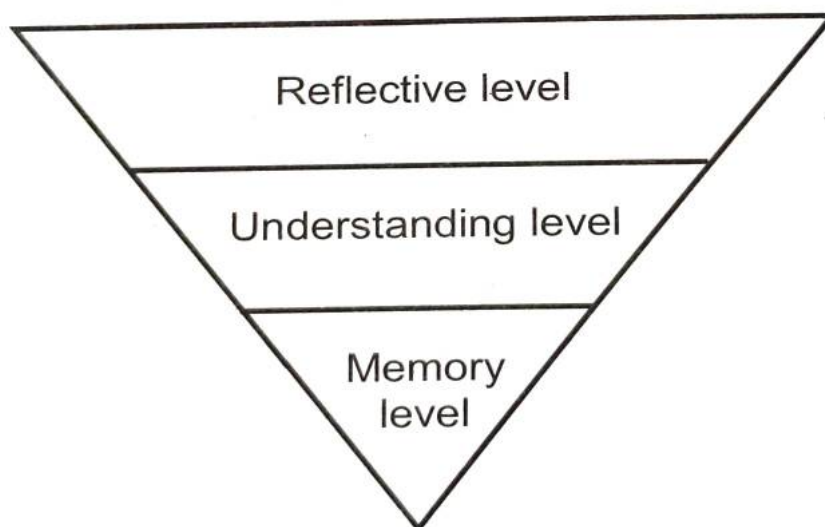
Organising of teaching-learning is the framework on which effective teaching is based. Careful and thoughtful planning allows instructional time to be maximised, standard

to be addressed, prior knowledge to be activated, misconceptions to be controlled and the diverse characteristics and learning needs to be considered. Effective teachers use different methods for teaching to teach knowledge. In this topic we are going to see three different methods for teaching.

3.4.1 Memory level of Teaching (Herbatian Model)

Teaching is a part of teaching-learning process. It is required to bring specific changes in a person according to the need of his society. It should improve the learning skills of students. Teaching has to bring changes in the behaviour of students.

Levels of Teaching



Memory Levels of Teaching

Objectives

1. The objective of the memory level teaching is just to impart information or knowledge to the learners.
2. Retention and reproduction through recall or recognition.

In this level of teaching content, presentation and its exercise is important for learning. Exphasis is given on the presentation of information and facts in a systematic manner

with a view to develop memory of students. At this level recall and recognition has important place.

Professor Herbart described the following five steps for the memory level of teaching. They are

- i. Focus
- ii. Syntax
- iii. Social system
- iv. Support system
- v. Evaluation system

We will discuss the above steps in detail.

i. Focus

Here importance is given on cramming of facts and development of following capacities.

- Training of mental aspects
- Providing knowledge of facts
- Retaining the learnt facts
- Recalling and Representing the learnt content

ii. Syntax

Herbert's five steps are

1. Preparation :

Important function of this step is to bring relevant ideas of the subject matter. The teacher plans about presentation of the contents.

2. Presentation :

The students are given new knowledge by testing their previous knowledge.

3. Comparison :

In this level teacher attempts to compare old facts and new facts, its similarities and dissimilarities, relationship between two subjects etc.

1. Generalization :

Pupils are given opportunities to think the content. Pupils formulate some principles and laws which can be used in the future life situations.

2. Application :

Teacher generate new situations to use learnt knowledge in new situation.

iii. Support System

Here supportive devise are used to make learning more effective. The Audio visual aids are used to present the content in an observable form.

iv. Social System

This system consist of authoritarian behaviour. The teacher is more active to construct and present the subject matter systematically and logically in the classroom. The learners are passive listeners and they have to memorize and reproduce the content and their activities are directed by the teacher. At this level teacher provides motivation through verbal praise or punishment. If the learner is motivated or reinforced continuously, then they learn better.

v. Evaluation system

Evaluation is the evidences about the realization of goals of teaching. Usually oral tests are given to measure the learnt material. The objective tests of recall and recognition types are given to evaluate the knowledge of the student.

Features of Memory Level

- Analysis of the subject matter is more important in this level of teaching. Much emphasis is given on the presentation of subject matter and not the experience of the students.
- Definite and systematic orders of subject matter is presented in this level of teaching.
- Teachers are not very flexible in the presentation of subject matter.
- The Mechanical handling of the various segments of content.
- Learning outcomes is of short duration.
- Interaction between teacher and students is at the lowest level.

Role of Teacher

- Dominating, active and authoritarian role
- He is the authority to instruct, deliver, direct, control and evaluate the performance.
- Takes initiatives, present the subject matter, directs the learner to do the activity in the desired manner.
- Right from the selection of the subject manner to its evaluation, the key process of teaching-learning remains with the teacher.

Suggestions for Memory Level Teaching

- Frequent recall or reproduction of content should be practised.
- Whole to part method should be used
- The content should have definite structure

- Teaching content should be integrated
- Contents should be presented systematically.
- Continuous reinforcement should be provided to the students.
- Students retention can be increased by practice.

Merits of Memory Level Teaching

1. Useful for children at lower classes.
2. The role of the teacher is important in this level of teaching and he is free to make choices of subject matter, plan it and can present it at will.
3. The knowledge acquired at memory level teaching forms a basis for the future.
4. Memory level teaching acts as the first step for understanding and reflective levels of teaching. It is pre-requisite for understanding level teaching.
5. It is useful for slow learners

Demerits of Memory Level Teaching

1. This does not contribute to the development of the students' capabilities.
2. At this level students learns by rote, so the knowledge gained does not prove helpful in real life situations as it does not develops the talents of students.
3. The pupils are kept in strict discipline.
4. Intelligence does not carry any importance in this type of teaching and it lacks motivation.
5. No interaction in the class
6. There is no guarantee of good retention and appropriate reproduction of the memorized material by the learner.

3.4.2 Understanding Level of Teaching (Morrison Model)

The teaching at the understanding level is of a higher quality than the one of the memory level. It is more useful and thoughtful from the point of view of mental capabilities. At this level of teaching, the teacher explains the student about the relationship between principles and facts and teach them how the principles can be applied. This method enables students to have complete command over subject material. No Cramming is encouraged here. The term understanding means to comprehend, to grasp, and to have knowledge of, to learn, to interpret and to infer etc.

Henry C Morrison clearly stated that understanding is not merely being able to recall something. It is not mere generalization deducted from specific facts; it is an insight into how it may be used in future situations. Morrison asserted that the outcome of all teaching is 'Mastery' and not memorization of facts. Morrison is the main promoter of understanding level of teaching.

Objectives :

1. Acquisition of required knowledge / facts
2. To see the relationship between acquired facts

Morrison has described five steps in his teaching model. They are

1. Exploration
2. Presentation
3. Assimilation
4. Organisation
5. Recitation

Let us discuss the above steps in detail

1. Exploration

In this step, the teacher tries to know the students 'entry-level behaviour' (Previous knowledge)

2. Presentation

The overview of the subject matter is given and the structure of the whole unit is discussed with the students.

3. Assimilation

In this step, the students study the subject matter deeply and try to related presented material to their internal self.

4. Organisation

The student present the acquired knowledge in a systematic manner without the help of teacher.

5. Recitation

This step is related to the students' verbal expression of the subject matter which is learned by them. They visit the whole content.

Morrison's Instructional Pattern

Morrison's teaching plan involves the following steps

1. Pretest
2. Teaching
3. Testing the result of instruction.
4. Changing the instruction procedures and
5. Teaching and testing again until the unit has been completely mastered by the student.

Subject matter and its presentation

The subject matter for the understanding level teaching is quite structured in terms of planning sequential and organised presentation and meaningful learning. It needs to

be linked with the previous knowledge of the students and with its utilization for acquiring new facts and application to practical life.

Role of the teacher

Like memory level of teaching, the teacher plays a dominant and authoritarian role at the understanding level of teaching. This type of teaching is also subject centred and hence teacher has to pay his full attention in presenting the subject matter to his students to attain the desired understanding objectives besides the knowledge objective. He has to take care of the fact that the students gain complete mastery over the subject matter in term of its full understanding and generalized insight rather than its mere memorization. Lecture, Demonstration or explanation, question and answer method, inductive followed by deductive approach and analytic method with synthetic approach are all carried out at the understanding level.

Evaluation

Well-planned oral written and practical examinations are conducted in and outside the classroom. However emphasis is given to test to know ability to analyse, synthesize and apply the generalised principles for deducing new facts and solving novel problems.

Merits of Understanding Level of Teaching

1. At this level of teaching students to make use of their thinking abilities.
2. Knowledge acquired at this level forms the basis of the reflective level of teaching
3. Here the teacher present subject matter before the students in an organized and sequential form. The new knowledge acquired is related to the previously acquired knowledge.

4. Here the students do not learn by rote. They learn by understanding the facts and information and their use and purpose.

Demerits of Understanding Level of Teaching

1. Teaching at this level is subject centred. There is no interaction between the teacher and students at this level.
2. In this type of teaching mastery is emphasized.
3. Ignore higher cognitive abilities.

3.4.3 Reflective Level of Teaching (Bigge and Hunt Model)

This level of teaching is the highest level of teaching - learning activity. Reflective level of teaching means, problem centred teaching. Hunt is the main promoter of reflective level of teaching. Reflective level teaching does not depend upon the memorization, understanding of concepts, and their application only; it demands the use of higher mental processes as

- Reasoning
- Thinking
- Analysing concepts
- Imagination
- ideas and thoughts

The learner will develop new insight to

- solve problems
- to discover
- to present generalization in the light of the new evidence
- to explore the field of knowledge

- to test and retest the achieved facts.
- This level basically involves the use of the scientific method.

Objectives

3. To develop insight into solve problems
4. To develop rational and critical thinking
5. To develop the ability of independent thinking and decision making.

Model of Reflective level of Teaching

Hunt developed the following steps for the Reflective level of Teaching. They are

a. Focus

To develop problem solving, critical and constructive, independent, original thinking

b. Syntax

Steps

- i. Creating a problematic situation
- ii. Formulation of Hypothesis
- iii. Verify hypothesis
- iv. Collection of data
- v. Testing of hypothesis

c. Social system

Student occupies the primary place and teacher secondary place.

d. Support system

Objective type test is not used but essay type test is used. Attitude, beliefs and involvement is evaluated.

Features of Reflective level of teaching

- It is the highest level of teaching.
- It is problem - centric approach of teaching.
- The students are assumed to adopt some sort of research approach to solve the problem
- Classroom environment is independent. The learners are self-motivated and active.
- It leads to the development of creative skills.
- The students can solve problems of their lives by reasoning logic and imagination and lead successful and happy lives.
- Essay type test is used for evaluation. Attitude, belief and involvement are also evaluated.
- clear goals for pupil learning

Methods used for Reflective level of teaching

- Problem-solving method
- Investigating projects
- Heuristic method
- Experimental method
- Inquiry oriented method
- Analytic method

Merits of reflective level teaching

1. The teaching at this level is not teacher - centred or subject-centred. It is learner-centred.
2. There is an interaction between the teacher and the student at the reflective level teaching.
3. At this level, teaching is appropriate for the higher class.

4. At this level, teaching is highly thoughtful and useful then the teaching at the memory or understanding level.
5. This method is very useful for gifted children.

Demerits of reflective level teaching

1. It is not suitable for small children. It is suitable only for mentally matured children.
2. At this level, the study material is neither organised nor pre-planned. Therefore students cannot acquire systemic and organised knowledge of their courses.
3. It is a time consuming process.
4. It is not suitable for Indian class room.
5. It is not applicable for dull students.

3.5. Designing a unit plan

The curriculum/Syllabus framers would organize the content matter for a specific subject/class into meaningful major divisions in terms of units. It may also be termed as chapters in book forms. It may have sub units or subdivisions these sub units may have a number of lessons with specific topical coverage. Thus a unit is an aggregation of several lessons, with specific core or central concept/idea. In text books these may be indicated as units or chapters.

For example the unit work, power and energy can be divided into four parts as

1. Respiration -Definition
2. Respiration- Animal respiration, plant respiration
3. Write about Human respiratory system.
4. Write about plant respiration

Definition:

Preston defines a unit as, "large block of related subject matter as can be over viewed by a learner."

Samford defines as, "A unit is an outline of carefully selected subject matter which has been isolated because of its relationship to pupil's need and interests."

Characteristics of a Good unit

- A unit should be meaningful segments of well-organized subject matter.
- General and specific instructional objectives should be clearly defined in the unit plan.
- A unit should consist of familiar and related subject.
- It should give new experience and knowledge to students.
- It should help the individual differences in learning.
- A unit should not be too lengthily or too short.
- The length of the unit should be such as to retain the interest of the students.
- It should give new experience to the students.

Unit planning

A unit plan involves planning of teaching a unit, teaching methods, evaluation of teaching activities, diagnosing and remedial steps all together is called unit planning.

Proforma of a unit plan

Name of the Teacher :

Subject : Biological Science

Standard :

Unit :

Class and section :

Total no. of periods :

Sl. No.	Subject – Units	Content analysis	Objectives and specifications	Teaching and learning activities	Teaching aids used	Evaluation

Advantages of unit planning

1. Since we are dividing the unit into small parts and teach to the students, the students are able to understand the subject matter clearly.
2. Unit plan helps the teacher to teach the students in systematic manner.
3. Unit plan help the students to understand the relationship between the facts and principles that make up the unit.
4. Unit plan help us to study all information pertaining the unit.
5. Unit plan helps to teach according to individual difference of students.
6. It helps the teachers to prepare teaching aids.
7. It helps to develop self confidence in students.

Steps followed to perform unit plan

The following steps have to be followed for a unit teaching - learning and activities.

Analysis of content

General instructional objectives

Specific instructional objectives

4. Planning the learning activities
5. Evaluation procedure
6. Remedial measures

1. Analysis of content

After selecting a unit, teacher has to analyse the contents of the unit in-depth. This helps the teacher to break up the unit into meaningful sub-units and retain the continuity throughout the unit.

2. General instructional objectives

Knowledge of content skill learnt is analysed here.

3. Specific instructional objectives

Specific instruction objectives of the content namely differentiating, establishing relationships, giving examples comparing, interpreting etc. are analysed.

4. Planning the learning activities

This part deals with the statements activities in teaching learning process keeping in mind individual difference, the psychology of learning and objectives. The teacher also has to plan suitable teaching methods that are to be employed here.

5. Evaluation procedure

Teacher has to evaluate with reference with reference to aims at the end of the unit or sub unit. Achievement test may be done and students' progress and the effectiveness of teaching strategies may evaluate.

6. Remedial measures

After analyzing the achievement test of the students, take remedial measures if the performance is not satisfactory.

Method of Performing unit plan

Biological science syllabus may be divided into small units. Each unit is to be divided into small parts/lessons and then decide the number of periods to be allotted for each lesson. Plan all the units of the syllabus have to be completed in time. According to content of the unit divide it into required number of small parts and allot periods.

Format to divide the syllabus into units

Class: 9

Subject : Biology

S.No	Chapter	Topic	No of Periods required	Note
1.	Human Anatomy and its functions	Introduction: Body and its parts	3	Diagnostic test
		Types of organs and its functions	7	
2.	Plant structure And functions	Introduction: Body and its parts	3	Achievement test
		Types of plants	7	
		Plant: structure its functions	2	

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Signature of the teacher

Biological syllabus is divided into units, each unit is divided into topics and lesson plan is prepared for the each topic and lessons should be completed within the time allotted.

Unit planning

A unit plan involves planning of teaching a unit, teaching methods, evaluation of teaching activities, diagnosing and remedial steps all together is called unit planning.

Content of a unit – Living World of plants – Plant Physiology

Sl. No.	Sub Division of the content	No. of periods required	Teaching Method	Resource Materials	Evaluation
1.	Introduction – Tropisms in Plants	1	Experiment method	Potted Plants 2 Nos.	The bending of root of a plant in response to water is called _____
2.	Types of Tropisms in Plants – Tropic Movements in plants – Nastic Movements in plants – Difference between them	2	Demonstration Method and Experiment Method	Bean seeds	Write the difference between tropic and nastic movements.
3.	Photosynthesis	1	Demonstration Method & Experiment method	Resources of Photosynthesis	The raw materials for photosynthesis are _____ and _____
4.	Transpiration	1	Experiment method	Potted Plant and Plastic bag	Transpiration takes place through _____ a) Fruit b) Seed c) Flower d) Stomato

3.5.1 Difference between lesson plan and unit plan

S. No.	Lesson plan	Unit plan
1	Planned for a small portion to give effective teaching in one period.	Planned to teach a large content continuously for 6 to 10 periods or more.
2	Lesson plan is partial.	Unit plan is complete.
3	Full concept in learning is not achieved.	Full concept in learning is achieved.
4	Teaching alone will take place.	Teaching, evaluation, remedial teaching will take place.
5	Questions will find a place in lesson plan to motivate self-thinking, language ability etc.	There are not questions in unit plan.
6	Assignments will find a place at the end.	There are no assignments.
7	Lesson plan is written for each period.	According to length of the unit, unit plan is written for 6 to 10 periods or more.
8	Method of teaching is not specified here.	Method of teaching is specified here.

3.6 Structure of a four-fold lesson plan

Introduction

We use the introduction to tell us a little about our lesson plan. Briefly describe the instructional techniques,

what students to learn and any activities or assessments that you think are particularly noteworthy. The title of the lesson plan should be concise, clear and descriptive. It should invite teachers to take a closer look at the plan. Remember that teachers may see only the title and a short abstract of our plan in a page of search results.

Objectives

We should give some general and specific objectives of the teaching lesson. These objectives should be written clearly to achieve the learning outcomes. Learning outcomes are what students are expected to learn after completing the lesson plan.

Learning outcomes should be closely related to the curriculum alignment but should not simply repeat goals and objectives of the standard course of study. Learning outcomes may be broader, address particular aspects of curriculum objectives or teach the curriculum in a special context. Each learning outcome should be clearly reflected in the activities and assessed at the conclusion of the lesson.

Session time

Session time is the amount of time a teacher will need to schedule for the lesson plan. We might specify minutes, hours, class periods.

Materials

Materials needed include resources used by both teacher and student, including books, handouts, teaching aids, paper and pencils, art supplies and so on.

Previous knowledge of Learners

Write the previous knowledge required to understand the lesson plan.

Procedures

We should follow the following procedures to take the lesson.

1. Motivation
2. Preparation
3. Presentation
4. Review/Recapitulation
5. Follow up/Assignments

1. Motivation

- It could be through interesting introduction.
- The essence of introduction consists of arousing the interest of the pupil by supplying a motive for the study of a new unit.
- It is therefore better to call motivation.
- The essence of motivation consists in making the pupil understand what they are about and feel the worth wellness of studying it.

The most desirable motivation is for the pupil to see purpose in what he is doing. He must

- Recognize their need and its significance
- Identify the cause of his difficulties
- Have confidence in his ability to correct the difficulties
- Be given the proper exercise materials, which provide self-evaluation, and
- Be provided with opportunities for application of skills they have learnt

2. Preparation

- This is for laying the basic foundation for construction of the new lesson.
- Test the previous knowledge of the students, so that they could understand the present topic based on those previous knowledge/concepts.

- If the pre requisite concepts are not clear, they may not understand the present concept.

3. Presentation

- The most important part of teaching all skills learnt through micro teaching must be applied here to make the teaching effective.
- Make thorough content analysis-concept by concepts and then present the concept from simple to complex, from easy to difficult etc.
- Provide various activities group work, demonstration, explanation, questioning and blackboard work.

4. Review

- Provide the summary of concepts taught in that period.
- It may be fast restating by teacher or by asking questions and eliciting answers from students.

This also will help to identify which part not understood well by students and brief explanation could be given.

5. Follow up

- It could be or simple assignment of asking to solve problems in exercise or
- Project work.

Students may be asked to observe from society or computer centers. It could be referring to nets/webs books etc.

The structure of four-fold lesson plan is given below.

Name of the Student Teacher	:	xxx
Name of the Guide Teacher	:	xxx
Name of the School	:	xxx
Class / Section	:	xxx

Subject : XXX
 Topic : XXX
 Sub Topic : XXX
 Period : XXX
 Duration : XXX
 No. of students in the class : XXX
 Date : XXX

Instructional objectives:

Write the list of objectives of the topic to be covered. i.e. Write what the student knows at lesson completion.

Instructional Resources Required:

Write the list of materials/aids required to explain the topic.

Previous Knowledge of learners:

Write the previous knowledge required to understand the new unit.

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Preparing and giving motivation to receive new knowledge / sequence of events that make up the lesson / independent practice - Recapitulation	Behavioural objectives are the smaller, observable, understandable and measurable. It must describe learning outcomes.	Learning experience refers to all interaction and practice in which learning takes place in classroom. In this column teacher / student activities will take place to attain the objectives.	On the basis of content / concept and learning experiences formative questions are framed here step by step.

Followup activities : Give homework and assignment in the topic covered.

Signature of Guide Teacher Signature of the Student Teacher

3.7 Preparation of a model lesson plans

Model Lesson Plan - 1

Name of the Student Teacher :	xxxxx
Name of the Guide Teacher :	xxxxx
Name of the School :	xxxxx
Class / Section :	X - A
Subject :	Biology
Topic :	Immune system
Sub Topic :	Diseases and causes
Period :	xxx
Duration :	45 minutes
No. of students in the class :	xxx
Date :	xxx

Learning objectives:

At the end of this Lesson students will be able to

1. tells about immune system
2. explains about immune system.
3. using his knowledge and understanding the students acquires the ability to solve questions in a new environment.
4. able to tell about the diseases and the reasons for it.
5. approaches the reasons for the diseases with research mentality

6. willing to solve the questions given to him.
7. able to differentiate the causative agents.
8. develops interest to knows about types of causative agents.
9. explain the categories of the tabular column.
10. solve the problem with interest

Teaching Aids:

Charts - Types of diseases caused by various agents and their life cycle

Models- Microorganisms

Immune system Diagram

Previous knowledge:

1. knowledge about the immune system
2. Knowledge about disease and its causes

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
Motivation	Recalls	Teacher show the images of micro organisms and check their previous knowledge.	Name the organisms on the table?
	Compares	Teacher asks questions. These structures acts as a vectors and spreads the diseases.	How it will spread the diseases?
	Recalls	These are called causative agents	How we can call these pathogens?
	Recalls	Bacteriophage	What are the names of the images placed on the table?

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
	Compares	Teacher asks questions. preventive measures are there for these images	How we can prevent these diseases?
Introduction of the lesson	Today we are going to see about immune system. Teacher writes the topic on the board.		
In a healthy state, a person keeps himself Biologically, mentally and mentally fit. The condition of the malfunctioning of the organ or system is called diseases.	Explains	Teacher explains about the immune system	What is meant by immune system?
	Compares	Teacher explains about the causative agents with the help of the model.	Define pathogens. Describe the life cycle of a pathogen?
Diseases caused by the organisms		Teacher ask questions by showing classification chart of disease causing micro organisms.	What are the types of pathogenic microbes?

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
	Microbes 1. Bacteria 2. Virus 3. Fungi 4. tapeworms		
	Gains knowledge	Teacher explains the structure of microbes and its infectious life cycle on the black board	Define infectious life cycle
Summarises the lesson by asking few questions from the important concept	Repeating the concept	1. What is meant by immune system? 2. Write the general symptoms found in a person with common cold?	

Signature of guide teacher

Signature of student teacher

Model Lesson Plan - 2

Name of the Student Teacher : xxxxx

Name of the Guide Teacher : xxxxx

Name of the School : xxxxx

Class / Section : VIII - B

Subject : Biology

Topic : Air, Water and Soil Pollution

Sub Topic : Air Pollution

Period : xxx

Duration : 45 minutes

No. of students in the class : xxx

Date : xxx

Learning objectives:

At the end of this Lesson students will be able to

1. tells about Air Pollution
2. explains about Air Pollution
3. using his knowledge and understanding the students acquires the ability to solve questions in a new environment.
4. willing to solve the questions given to him.
5. observes the environmental news about with research attitude.
6. wishes to prevent Air pollution.

Teaching Aids:

Charts,

Models

Previous knowledge:

1. About the layers of atmosphere.
2. About causative agents of pollution.

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
Motivation	Identifies	Teacher shows the picture about different layers of atmosphere.	How it will spread the diseases?
Atmosphere	Recalls	There are four layers in the Atmosphere. It has 20.9% oxygen,	What is the proportion of gases present in air?

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
		78% nitrogen, 0.03 carbon di oxide, neon, hydrogen, less amount of water vapour.	
Introduction of the lesson	Acquires knowledge	Teacher writes the topic on the black board	What is meant by Air pollution?
	Recalls	The change in Biological and chemical processes is called pollution.	
	Understands	Teacher calls the student and ask him What is natural sources?	What are the two main causative agents for air pollution?
Natural sources	Describes	Volcanic eruption, forest fire these are the reasons for natural air pollution.	What are the natural polluting agents?
Artificial sources	Explains	Smoke from the factory, vehicles, Aeroplanes and from the power stations.	What are the artificial polluting agents?
Air pollutants	Acquires knowledge	1. Carbon mono oxide 2. Carbon di oxide 3. Hydrogen oxide 4. Sulphur di oxide All these are found in the air.	What are the air pollutants found in air?

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
Effects of Air pollution	Explains	1. Acid rain 2. Ozone hole 3. Ozone – colourless gas, found in the stratosphere layer. Density of the ozone layer is reduced due to the pollutants found in the atmosphere, results in the ozone hole.	What are effects of Air pollution?
Controlling Air pollution Application	Acquires knowledge	Avoid using sludge oil 1. Fuel 2. Passive gasoline 3. Bio diesel 4. Compressed natural gases should be used. 5. Planting more trees to reduce the level of carbon di oxide. Students uses his knowledge in future to create awareness.	

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
Summarise the lesson by asking few questions from the important concept	Repeats	1. What is meant by air pollution? 2. What are the Air pollutants? 3. How to control Air pollution?	

Signature of guide teacher

Signature of student teacher

Model Lesson Plan - 3

Name of the Student Teacher : xxxxx

Name of the Guide Teacher : xxxxx

Name of the School : xxxxx

Class / Section : VI - C

Subject : Biology

Topic : Health and Hygiene

Sub Topic : Nutrients

Period : xxx

Duration : 45 minutes

No. of students in the class : xxx

Date : xxx

Learning objectives:

At the end of this Lesson students will be able to

1. tells the importance of health and hygiene.
2. explains the good and hygienic habits in our life.
3. evaluate the importance of nutrients present in our food.

4. List out the foods that contains carbohydrates, fats and protein.

Teaching Aids:

Charts,

Flash Cards

Previous knowledge:

1. Personal Hygiene
2. Knowledge of food items

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
Motivation	Identifies	Teacher ask students simple question about health care.	1. Why do we wash our hands before eating food? 2. Why do we brush our teeth daily?
Introduction of the Lesson Different components of food	Acquires knowledge	Teacher shows the chart that contains carbohydrates, fats, food items. Carbohydrates - Potatos, Rice Fats - Meat , Egg	1. Tell any two food items that contain carbo-hydrates 2. Tell any two food items that contain fats.
	Acquires knowledge	Carbohydrates are a large group of organic compounds occuring in foods in the form of sugar, starch and dietary fibres. It is the body's main source of energy.	What are Carbohydrates?

Content	Specification of the behaviour objectives	Learning objectives (Teacher/learner activities)	Evaluation
		Fat is also an energy giving food that give more energy than Carbohydrates. They insulate our body and protect our cells.	What is fat?
		Protein is an organic compound which are an essential part of all living organisms specillay for body tissues	What is protein?
	Recalls	Eggs, milk, Cheese, chicken, Beans etc. Beans - Protein Meat - Fat Rice - Carbo-hydrates etc.	Name few protein foods. Teacher use flash cards to identify the food components.
Health problems due to fat	Acquires knowledge	Fat around our mid section gives serious health problems. Excess fat sits right below the skin surface. It leads to heart disease, diabetes, cancer, pains etc.	What are the Health problems due to excess fat?
Summarises the lesson by asking few questions from the subject matter	Repeats	1. What are Carbohydrates? Tell any two food items that contain carbohydrates. 2. What are Fats? Tell any two food items which contain fats. 3. What are proteins? Tell any two food items which contain proteins. 4. What the diseases for excess fat?	

Signature of guide teacher

Signature of student teacher

Review Question

1. Define Lesson plan. Write the need and importance of Lesson Plan.
2. Explain the steps followed to prepare a lesson plan.
3. Explain in detail the Memory Level of Teaching. Write its Merits and Demerits.
4. Explain in detail the Understanding Level of Teaching. Write its Merits and Demerits.
5. Explain in detail the Reflective Level of Teaching. Write its Merits and Demerits.
6. Define a Unit Plan. What are the steps to be followed to perform a unit plan. Write its advantages.
7. Write the difference between Unit plan and Lesson plan.
8. Explain in detail the structure of a four-fold lesson plan.
9. Write a model lesson plan for IX Std. Biological Science.
10. Write a model lesson plan for VIII Std. Biological Science.